comparing results; intro to flare

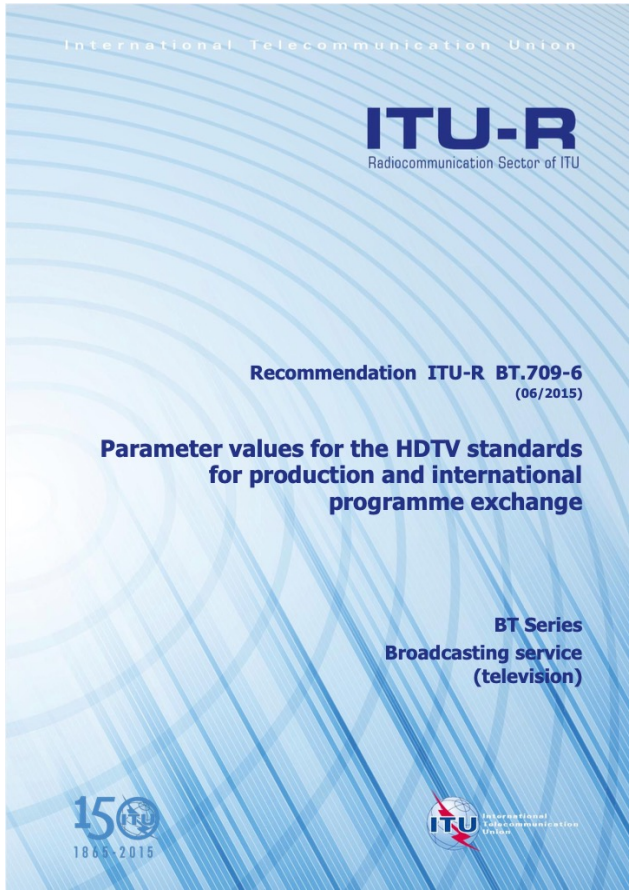
(PCM day 2 session 4)

What was the range of gamma values seen?

- Any patterns as to divergence of result?
- Which patches were the hardest to evaluate for closeness?

Revisiting BT.1886 and its context

- The offset term only accounts for what the display surface emits
- It doesn't account for what the display surface *reflects*



Annex 1

Reference electro-optical transfer function

The reference EOTF is specified by the equation:

$$L = a(\max[(V+b),0])^{\gamma}$$

Flare can be an issue

- Monitor hoods pretty good at keeping out direct spill light
- Multi-bounce light harder to eliminate
- Not just desktop displays: cinemas too
  - minimize spill from exit sign onto screen
  - dark seats and carpets
  - but: still have light-colored clothing

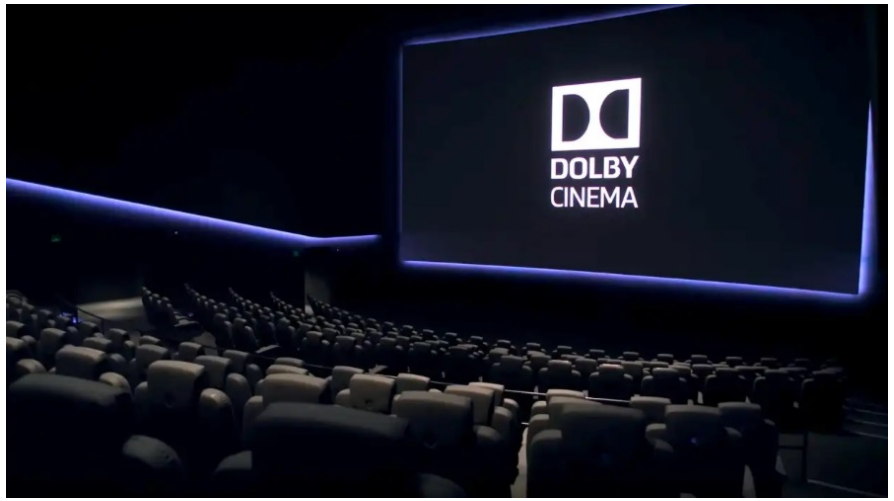
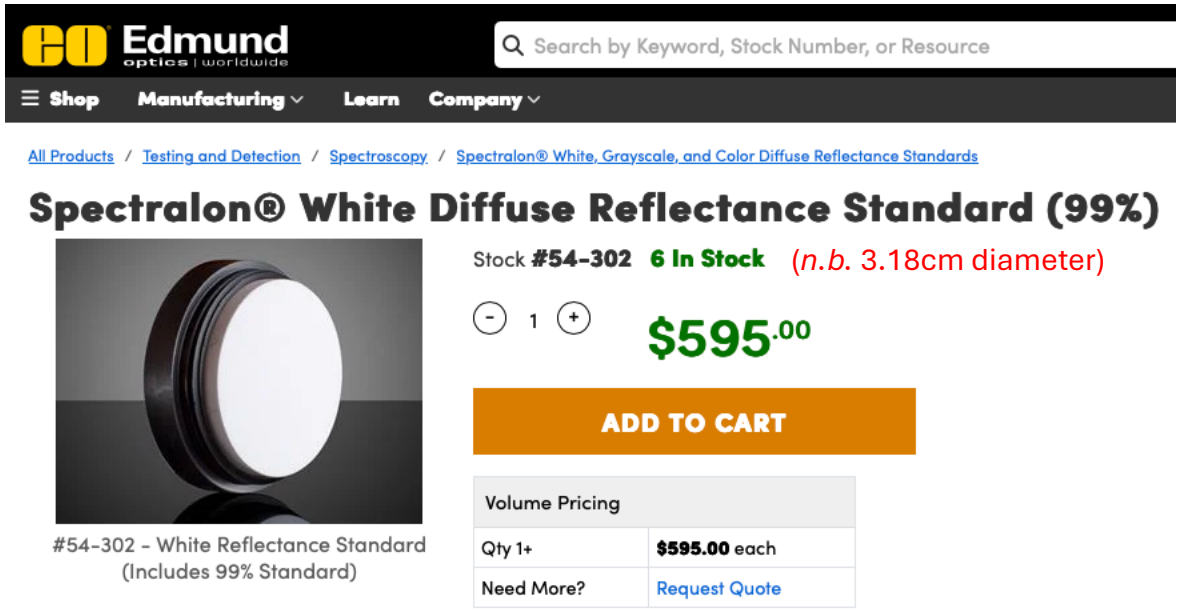


image sources: eizo.ch, nofilmschool.com

Measure flare with a PRD

- A PRD is a perfect reflecting diffuser
  - ideally spectrally neutral
  - ideally fairly efficient
  - would be nice if they were cheaper



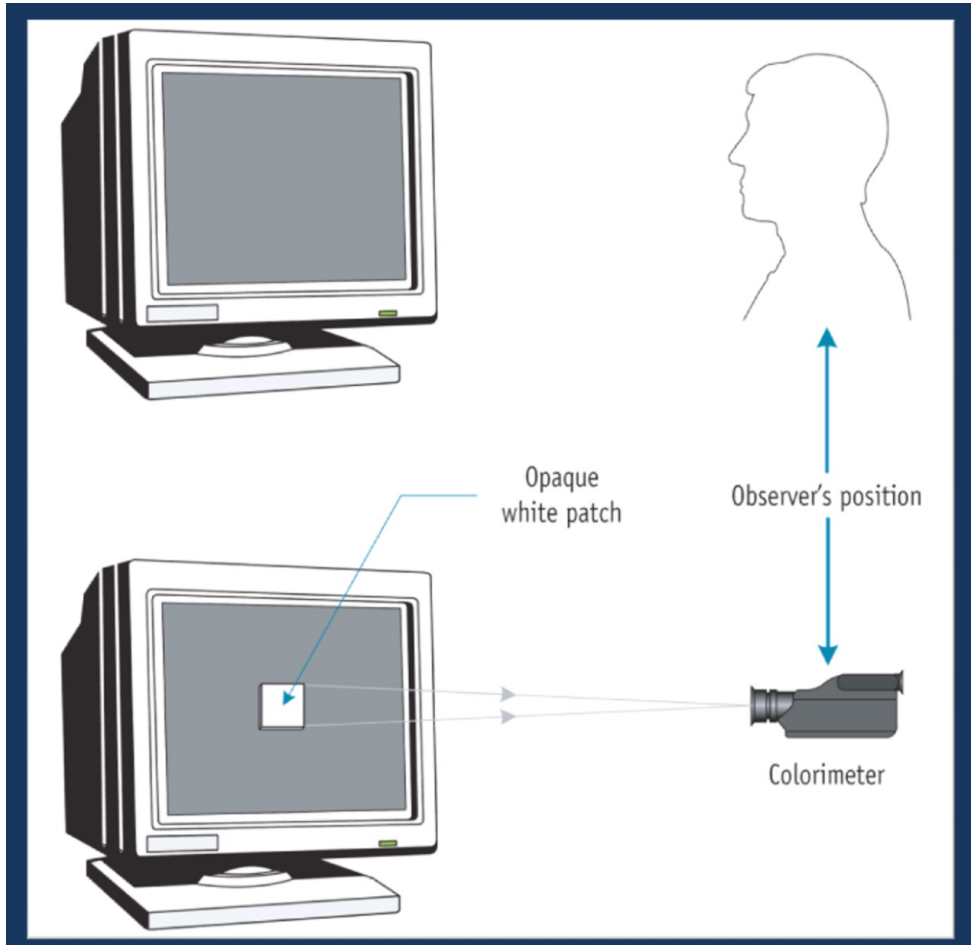


Figure 5.17 One method for measuring flare light reflected from the face of a CRT.

image sources: Edmund Optics catalog; G&M 2e

Better graphs for low-light

- What should we expect from the monitor?
  - ASUS PA278QV (*not* Gen2)
  - spec page
    - https://www.displayspecifications.com/en/model/3fef1e5b
    - static contrast 1,000:1
    - 8-bit panel
    - peak brightness 300 nits
  - Therefore black at 0.3 nits
  - sRGB mentioned 4X on spec page
  - Let's plot the EOTFs of a monitor designed to respond according to sRGB's description and for comparison, BT.1886

Why is sRGB so much brighter near black?

- sRGB was written by computer manufacturers for office environments with high ambient light levels

- BT.1886 was written by broadcasters, when they liked to think of people watching TV in their living room at night, with low ambient light levels

The paper, which you will find in your papers subdir, after the break, has more words than standards do, and is much more open to multiple interpretations

Proposal for a Standard Default Color Space for the Internet—sRGB

Matthew Anderson , Ricardo Motta†, Srinivasan Chandrasekar*, Michael Stokes†*
**Microsoft, where, where*
†Hewlett-Packard, Palo Alto, California

Introduction

HP and Microsoft propose the addition of support for a standard color space, sRGB, within the Microsoft OS's, HP products and the Internet. The aim of this color space is to complement the current color management strategies by enabling a third method of handling color in the OS's and the Internet that utilizes a simple and robust device independent color definition that will provide good quality and backwards compatibility with minimum transmission and system overhead. Based on a colorimetric RGB color space well suited to Cathode Ray Tube (CRT) monitors, television, scanners, digital cameras, and printing systems, such a space can be supported with minimum cost to software and hardware vendors. Our intent here is to promote its adoption by showing the benefits of supporting a standard color space, the suitability of the standard color space we are proposing, describe some of the system issues and propose a methodology for its implementation on the Web.

Part 1—Definition of the sRGB Color Space

Recently the International Color Consortium has proposed and is positioned to provide breakthrough solutions to problems in communicating color in open systems. Yet the ICC profile format does not provide a complete solution for all situations.

Currently, the ICC has one means of tracking and ensuring that a color is correctly mapped from the input to the output color space. This is done by attaching a profile for the input color space to the image in question. This is appropriate for high end users. However, there are a broad range of users that do not require this level of color quality, could easily and confidently communicate color without further color management overhead in the most common situations. The two major factors of this RGB space are the colorimetric RGB definition and the gamma value of 2.2, along with a number of secondary details necessary to enable the clear and unambiguous communication of color.

Colorimetric RGB

The dichotomy between the device dependent (e.g. amounts of ink expressed in CMYK or digitized video voltages expressed in RGB) and device independent color spaces (such as CIELAB or CIEXYZ) has created a performance burden on the system because of the complexity of the transforms, and a reliability gap since the complexity and variety of the transforms make it hard to ensure that the system is properly configured.

To address these concerns and serve the needs of PC and Web based color imaging systems, we propose a colorimetric RGB specification that is based on the average performance of personal computer displays. This solution is supported by the following observations:

- Most computer monitors are similar in their key color characteristics—the phosphor chromaticities (primaries) and transfer function
- RGB spaces are native to displays, scanners and digital cameras, which are the devices with the highest performance constraints
- RGB spaces can be made device independent in a straightforward way. They can also describe color gamuts that are large enough for all but a small number of applications.

source: Imaging, Science & Technology website (imaging.org)

Adding flare to sRGB and BT.1886

- 0.75% viewing flare
- 300 nit peak
- 2.25 nits flare!

source: Imaging, Science & Technology website (imaging.org)

We can compensate by subtracting but…

- …that wouldn’t help when the desired light level is less than the flare we would be subtracting
- …a 2009 textbook’s idea of ambient flare might be high

source: Imaging, Science & Technology website (imaging.org)

We subtract flare from the *reproduced* signal

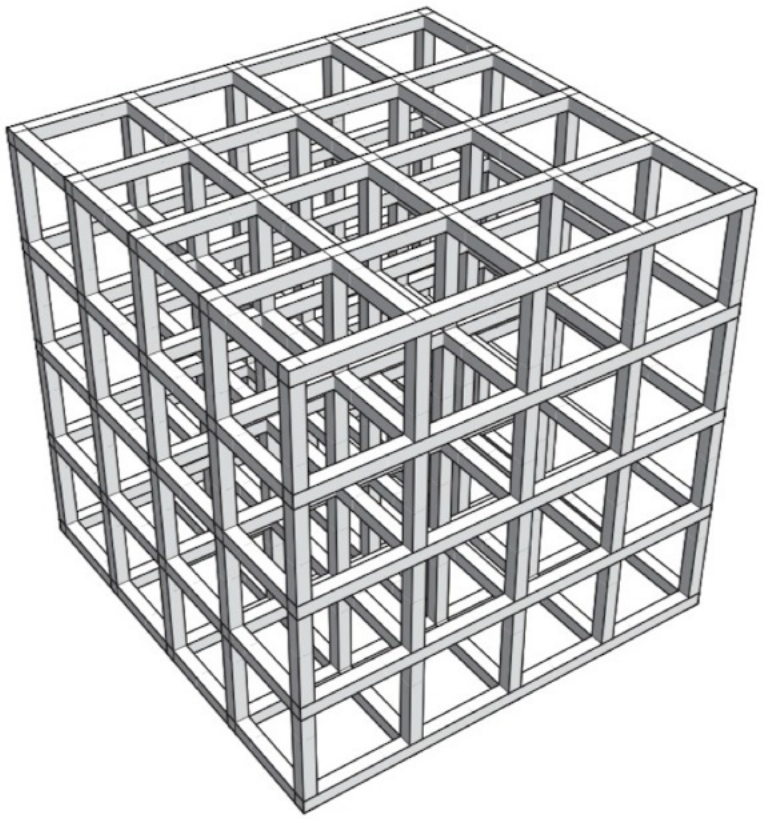
- Our target was to match (proportionately speaking) the scene exposure value, 1:1
- Flare interfered with that
- Determine what signal level after flare would give us the signal level we actually want
- Change the image content according to that determination

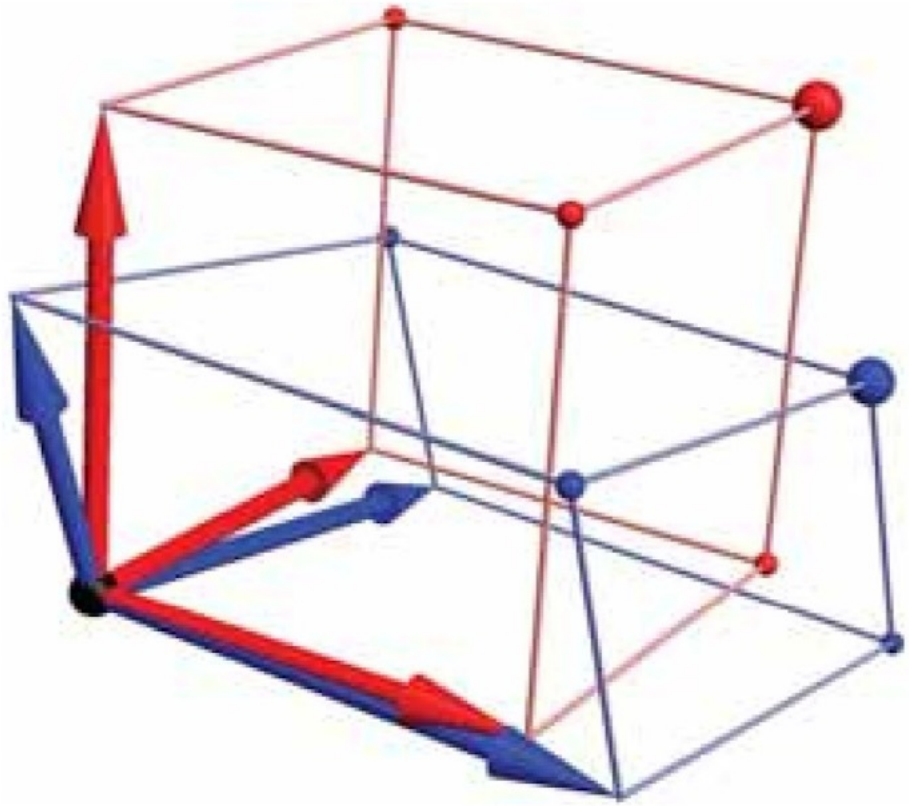
source: Imaging, Science & Technology website (imaging.org)

Enter Helen and John

- Early camera test for ALEXA 35
- Available as ARRIRAW MXF
  - scene-referred
  - can convert to other scene-referred color spaces
    - linear AWG4 (TIFF)
    - linear ACES AP0 (EXR)
  - can convert to display-referred spaces
    - display-ready Rec709 (TIFF)
    - display-ready P3-D65 (TIFF)

Primaries as a change of basis





start with ARRI Wide Gamut 4

ARRI LogC4

Logarithmic Color Space

SPECIFICATION

Date: 23rd January 2025

4.2 ARRI Wide Gamut 4

ARRI Wide Gamut 4 (AWG4) is the name of the color primaries used in ARRI LogC4. All chromaticity values are defined with CIE 1931 2 Degree Standard Observer Colorimetry.

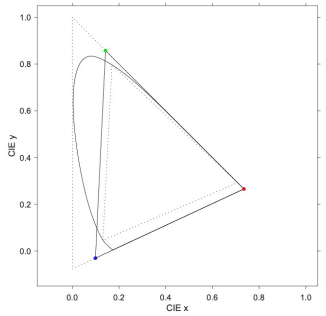


Figure 3: The 2D gamut bounded by ARRI Wide Gamut 4 in the CIE 1931 chromaticity diagram is shown as the solid line with colored markers. The larger and smaller gamuts with dotted lines are ACES AP0 (SMPTE ST 2065-1) and ITU-R BT.2020 respectively.

4.2.1 RGB Primaries

The RGB primaries of ARRI Wide Gamut 4, are defined as follows:

	CIE x	CIE y
Red	0.7347	0.2653
Green	0.1424	0.8576
Blue	0.0991	-0.0308

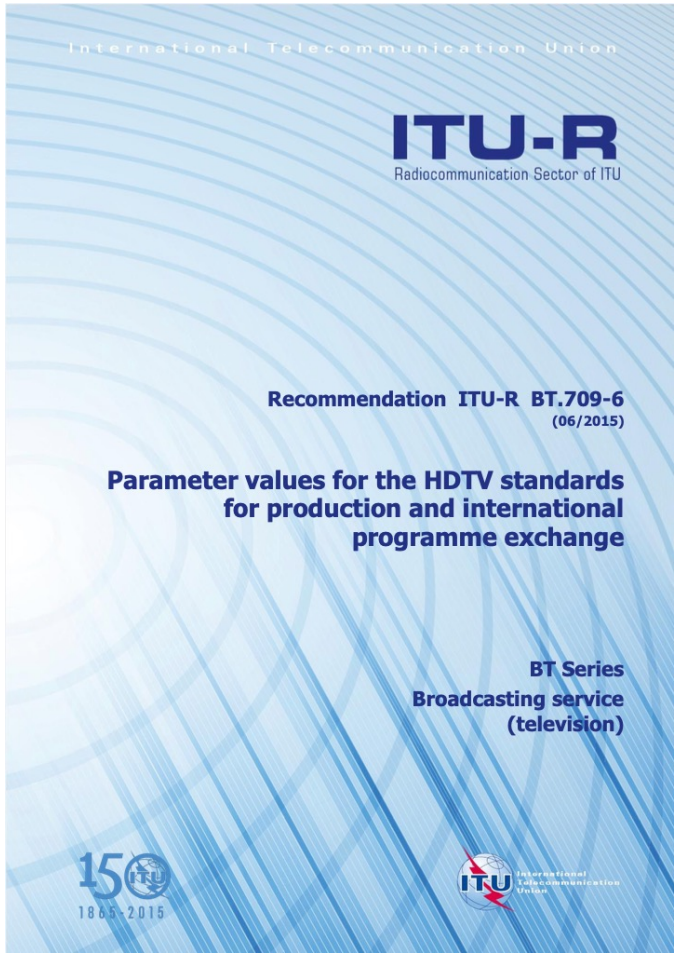
4.2.2 White Point

The White Point of ARRI Wide Gamut 4 is CIE Standard Illuminant D65, provided here for completeness:

	CIE x	CIE y
White	0.3127	0.3290

source: https://www.arri.com/resource/blob/278790/f3318e8c9c65617d8c5ca3f8b3e32051/2023-05-arri-logc4-specification-data.pdf

end with ITU-R BT.709 (even though sRGB)



1 Opto-electronic conversion

Item	Parameter	System Values	
1.1	Opto-electronic transfer characteristics before non-linear pre-correction	Assumed linear	
1.2	Overall opto-electronic transfer characteristics at source[1]	$V = 1.099\ L^{0.45} - 0.099$ for $1 \geq L \geq 0.018$ $V = 4.500\ L$ for $0.018 > L \geq 0$ where: L : luminance of the image $0 \leq L \leq 1$ V : corresponding electrical *signal*	
1.3	Chromaticity coordinates (CIE, 1931)	x	y
	Primary – Red (R) – Green (G) – Blue (B)	0.640 0.300 0.150	0.330 0.600 0.060
1.4	Assumed chromaticity for equal primary signals (Reference white)	D_{65}	
		x	y
	$E_R = E_G = E_B$	0.3127	0.3290

(1) In typical production practice the encoding function of image sources is adjusted so that the final picture has the desired look, as viewed on a reference monitor having the reference decoding function of Recommendation ITU-R BT.1886, in the reference viewing environment defined in Recommendation ITU-R BT.2035.

2 Picture characteristics

Item	Parameter	System Values
2.1	Aspect ratio	16:9
2.2	Samples per active line	1 920
2.3	Sampling lattice	Orthogonal
2.4	Active lines per picture	1 080
2.5	Pixel aspect ratio	1:1 (square pixels)

source: https://www.itu.int/rec/R-REC-BT.709-6-201506-I/en

Just carefully copy down all those chromaticities…

4.2.1 RGB Primaries

The RGB primaries of ARRI Wide Gamut 4, are defined as follows:

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		x	y
	$E_R = E_G = E_B$	0.3127	0.3290

source: https://www.itu.int/rec/R-REC-BT.709-6-201506-I/en

...and then go through this math...

ARCHIVED 2010

REAFFIRMED 2002

SMPTE RECOMMENDED PRACTICE

RP 177-1993

Derivation of Basic Television Color Equations

1 Scope

This practice is intended to define the numerical procedures for deriving basic color equations for color television and other systems using additive display devices. These equations are first, the normalized reference primary matrix which defines the relationship between RGB signals and CIE tristimulus values XYZ; then, the system luminance equation; and finally, the color primary transformation matrix for transforming signals from one set of reference primaries to another set of reference primaries or to a set of display primaries.

2 Normative reference

The following standard contains provisions which, through reference in this text, constitute provisions of this practice. At the time of publication, the edition indicated was valid. All standards are subject to revision, and parties to agreements based on this practice are encouraged to investigate the possibility of applying the most recent edition of the standard indicated below.

CIE Publication 15.2 (1986), Colorimetry

3 Reference primary matrix

3.1 Input data

3.1.1 For television systems, reference white for both reference camera and display is CIE illuminant D_{65}. The chromaticities of D_{65}, rounded to four significant digits (CIE 15.2), are:

$x = 0.3127$
$y = 0.3290$

Other displays may utilize other white points. The CIE coordinates of some other standard CIE illuminants are:

	x	y
D_{55}	0.3324	0.3474
D_{50}	0.3457	0.3585
Ill C	0.3101	0.3162

(A more complete discussion of reference white and operating white is given in annex A.)

3.1.2 The chromaticity coordinates of the system reference primaries or display primaries shall be specified to a minimum of three significant digits using the 1931 CIE system of colorimetry (x, y coordinates). More precision may be used in the starting data if available. If the primary chromaticities are provided in the CIE 1976 uniform chromaticity coordinate systems (u′,v′), then they must be transformed to the 1931 CIE x, y chromaticity coordinates using the transformation given below (see annex D (1)).

$$x = \frac{9u'}{12 + 6u' - 16v'}$$

$$y = \frac{4v'}{12 + 6u' - 16v'}$$

Should the primary chromaticities be provided in the obsolete 1960 UCS system (u,v), these values should first be transformed to the 1976 system and then to the 1931 x,y values.

$$u' = u; \quad v' = 1.5\,v.$$

3.1.3 In the event that starting data are not available as chromaticity coordinates for a white point or display primaries, they should be computed from the spectral power distributions using the colorimetric integration tables and procedures given in CIE 15.2.

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595 W. Hartsdale Ave., White Plains, NY 10607
(914) 761-1100

Approved November 1,1993

3.2 Output data

The numerical coefficients of the output matrix must be computed accurately to four decimal digits. To avoid effects of rounding and truncation errors and to ensure this accuracy in the final result, it is recommended that all computations be carried out to 10-digit accuracy with the output matrix being derived directly from the chromaticity coordinates of the RGB primaries and reference white with no rounding or truncation of intermediate results. The final 10-digit result is rounded to the required 4-digit accuracy. A set of example computations accurate to 10 digits is given in annex B to aid in checking computational equipment.

3.3 General procedure

The general procedure for deriving the matrix relating normalized linear RGB signals to CIE XYZ tristimulus values is described in this clause and an example derivation is given in annex B. The RGB signals are normalized such that reference white has the values R=G=B=1.0. The step-by-step process is as follows:

3.3.1 Obtain the CIE x y chromaticity coordinates of the reference white (D_{65} for television) and of the RGB primaries.

3.3.2 Compute the z coordinate for the reference white and each of the RGB primaries:

$$z = 1 - (x + y)$$

3.3.3 Form the following matrix and column vector from the x y z numerical values of the reference primaries and white:

$$P = \begin{bmatrix} x_R & x_G & x_B \\ y_R & y_G & y_B \\ z_R & z_G & z_B \end{bmatrix} \qquad W = \begin{bmatrix} x_W / y_W \\ 1 \\ z_W / y_W \end{bmatrix}$$

Note that the W vector, representing the reference white, has been normalized so that white has a luminance factor of 1.0; i.e., Y = 1.0. This is necessary so as to cause the video reference white signal (R=G=B=1) to produce the reference white with a unity luminance factor.

3.3.4 Compute the coefficients C_i on the left side of the equation below by multiplying the W vector by the inverse of the P matrix. Note the notation P^{-1} indicates the matrix inversion operation. These coefficients are normalization factors which normalize the units of the RGB primaries such that a unit amount of each combine to produce the white point chromaticities with a luminance factor of 1:

$$\begin{bmatrix} C_R \\ C_G \\ C_B \end{bmatrix} = P^{-1} \bullet W$$

3.3.5 Form the diagonal matrix from the coefficients C_i computed in 3.3.4:

$$C = \begin{bmatrix} C_R & 0 & 0 \\ 0 & C_G & 0 \\ 0 & 0 & C_B \end{bmatrix}$$

3.3.6 Compute the final normalized primary matrix NPM as the product of the P and C matrices:

$$NPM = \begin{bmatrix} X_R & X_G & X_B \\ Y_R & Y_G & Y_B \\ Z_R & Z_G & Z_B \end{bmatrix} = P \bullet C$$

3.3.7 This matrix, NPM, is the final result and relates television linear RGB signals to CIE XYZ tristimulus values as follows:

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} X_R & X_G & X_B \\ Y_R & Y_G & Y_B \\ Z_R & Z_G & Z_B \end{bmatrix} \bullet \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

3.3.8 The luminance equation for this set of primaries is the second row of the NPM matrix:

$$Y = Y_R\,(R) + Y_G\,(G) + Y_B\,(B)$$

In some cases, the NPM matrix values rounded to four digits may result in a luminance equation whose terms do not sum to 1.0. In that situation, the NPM matrix should be column normalized to force the second row to sum to 1.0.

3.3.9 Computations of color-difference signal coefficients should use all 10 digits of the luminance equation as determined above. These data should be multiplied by applicable scaling factors before rounding. Round to four decimal places

source: SMPTE

…or just use Colour so we can focus on flare again

- Take a look at 02-04-convert_helen_and_john.ipynb
- This uses:
  - Colour which already has these matrices defined for us
  - OpenImageIO to read and write the image
  - NumPy to apply the matrices to the pixels
- Even with half of it being error-checking code, it’s only 45 lines of code
- The result is a scene-linear image expressed in the primaries and white point that both BT.1886 and sRGB monitors expect

Is flare subtraction actually taking place?

- Open 02-04-flare_removal.ipynb and run it
- The result is not dramatic
- Now change the flare to be removed, multiply it by 10
  - color management image changes are often subtle
    - but so are bugs where code is silently skipped
  - wildly inappropriate values are a good way to test this sort of thing

Note how lifeless the images look

- Perfectly proportional reproduction looks bad
- Coming up: psychophysical effects
- Questions?
- Break